An Efficient Annotation Method for Big Data Sets of High-Resolution Earth Observation Images

Zeshan Lu[1], Kun Liu[1,2,*], Zhen Liu[1,*], Cong Wang[3], Maoxin Shen[1] and Tao Xu[1,2]

1 School of Information Science and Engineering, University of Jinan, Jinan 250022, China
2 Shandong Provincial Key Laboratory of Network Based Intelligent Computing, Jinan 250022, China
3 Lianyuan NO.1 School, Lianyuan, Hunan 417100, China
liukun@ujn.edu.cn; liuzhen@ujn.edu.cn

ABSTRACT

High-resolution earth observation images have increased dramatically because of the increasing of remote sensing satellites. Researchers must do large-scale target annotations to meet the training needs of deep neural network based model. However, most existing datasets contain an insufficient number of annotated samples, due to the inefficient manual annotation process which reason lies in the large number of remote sensing images, huge size, numerous targets, and high accuracy requirements. This paper proposed an efficient annotation method for big data sets of high-resolution earth observation images, in which the annotation process is divided into two parallel sub-processes, fast panchromatic image labeling and multi-spectral image fusion. Automatic scale transform is utilized for annotation of fused imagery. Experimental results show that the proposed method could improve the accuracy and efficiency of target labeling. Mask-RCNN and Faster-RCNN based target detection results demonstrate the validity of the big dataset annotated via our method.

CCS Concepts

• Information systems→Information systems applications→Spatial-temporal systems→Geographic information systems.

Keywords

Annotation method; high-resolution images; Mask-RCNN; target detection.

1. INTRODUCTION

In recent years, high-resolution remote sensing images have been widely used in government management [1], people's lives [2] and military affairs. As a key step, target detection could provide location of region of interest and change detection results. Large-scale annotated datasets play a very important role in improving the performance of target detection [3] methods, the reason is that the target detection algorithm based on deep neural network (DNN) has become the commonly used algorithm in remote sensing target detection in recent years.

However, training stage of DNN is a supervised learning process, which requires a large number of samples and accurate labeling of samples. For example, building detection in remote sensing images requires manual labeling of a large number of various building samples, and accurate labeling boundary adjustment is must be made for irregular buildings. However, the traditional method of target labeling in remote sensing images is to organize all stages of work in series, which brings difficulties to manual labeling of high-density and large-scale remote sensing targets.

Considering the building labeling of GF-2 satellite images, each step of the traditional annotation process is grouped in series. Firstly, the fusion process of panchromatic image and multispectral image takes about 500 seconds, and fused image reaches 6GB. Skilled labeler will spend about 10 hours labeling the buildings it contains. Researchers cannot annotate during fusion stage. The resolution of fused images decreases, resulting in lower accuracy of annotation. Aiming at the above problem, this paper proposes an efficient target annotation method for big data sets of high-resolution remote sensing images, in which each step of labeling process is grouped in parallel. The proposed method could support efficient and accurate target labeling for large number of buildings in GF-2 satellite images. Mask-RCNN [4] and Faster-RCNN detection methods are utilized in experiments for validation of labeled targets.

This paper has two main contributions: (1) an efficient labeling method for high-resolution images data set is proposed, which is used to build the largest GF-buildings dataset. (2) Mask-RCNN and Faster-RCNN were used to verify the validity of the dataset, and satisfactory experimental results were obtained. The organization of rest paper is as follows: Section 2 introduces the related work. Section 3 proposed the efficient labeling method and section 4 shows the experimental results with analysis, the conclusions was given in section 5.

2. RELATED WORK

For target detection of remote sensing images, a number of high-resolution remote sensing datasets such as SIRI-WHU Dataset [5], WHU-RS19 Dataset [6], RSC11 Dataset [7] and UC Merced Land-Use Dataset [8] have been proposed. But the number of samples of these data sets is relatively small, and most of samples were selected from Google Earth or the United States Geological Survey (USGS) National Map. The image number of the SIRI-WHU Dataset, WHU-RS19 Dataset, RSC11 Dataset and the UC Merced Land-Use Dataset are 2400, 1005, 1232 and 2100 respectively, demonstrating that the data collation is complex and prolix, which lead annotation process takes more time and effort.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.
ICBDT2019, August 28–30, 2019, Jinan, China
© 2019 Association for Computing Machinery.
ACM ISBN 978-1-4503-7192-6/19/08…$15.00
https://doi.org/10.1145/3358528.3358566

At present, the unified and meticulous high resolution images data annotation method is lacked. To establish a suitable dataset for maintaining the needs of various research field, it is necessary to label the interested objects constantly and set attribute parameter, which is time-consuming. High efficiency annotation method proposed in this paper is universal and can be used on most high-resolution Earth observation data. Deep learning makes the machine deal with problems like humans which has great achievements in image detection and classification detection, etc. For example, Tsung-Yi Li et al. developed a top-down architecture with horizontal connections for constructing a feature pyramid network (FPN) architecture [9] of high-level semantic feature graphs at all scales, improving the accuracy of target detection. Region-based convolutional neural networks(R-CNN) are widely used in the field of target detection, so we chose the popular Mask-RCNN and Faster-RCNN algorithm for experiments.

3. ANNOTATION METHOD

The annotation process we proposed need marked on the high resolution panchromatic images at first. The annotation result can be mapped to multispectral and fused images. So it is no longer necessary to re-mark from multispectral or fused images to improve the efficiency of labeled which can save time. The annotation process is shown in figure 1.

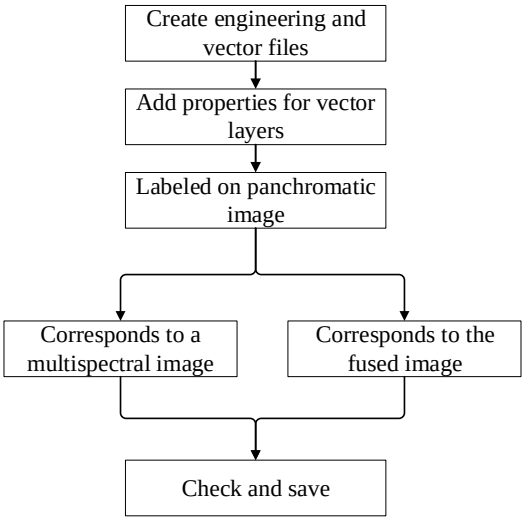


Figure 1. Annotation flow char.

3.1 Image interception

Labeling efficiently, reducing unnecessary time consuming, and making the data utilization of the labeled data as high as possible needs many steps. First, image samples (take artificial buildings as an example) with great feature of rich, dense, clear and sized should be intercepted from the original high-resolution remote sensing image. Intercepted images are shown in figure 2.

3.2 Annotation process

3.2.1 Create mxd engineering and vector files

An mxd file and a vector file to store the annotation information should be created at first. The coordinate system must be consistent with the geographic coordinate system of the imported image when creating the shp file, if the geographic coordinates are inconsistent, the annotation information in the shp file cannot be displayed on the image, which affects the follow-up work.

Then import the images need to annotate. The images we used here are high-resolution remote sensing images selected from gaofen-2 remote sensing images with little cloud cover, relatively clear coverage and rich objects. And select the part with large number of objects, clear edges and little irrelevant information to intercept with a rectangle. The image size will be limited between 2000*2000 pixels and 5000*5000 pixels (all vector files should have the same geographic coordinates as the original image)

3.2.2 Vector layer adds attributes

Some attributes need to be added on the vector layer to make labeling more convenient. A type column on the layer property sheet should be added and enable editing. The rectangle tool in the editor used to draw the initial template in the blank space of the image, and equal numbers small rectangles are drawn to indicate how many scene categories have to be set. In this paper, artificial buildings are divided into housing construction, factory building, circular building and other building, the initial template has four small rectangles, and these four categories need to fill in the type column.

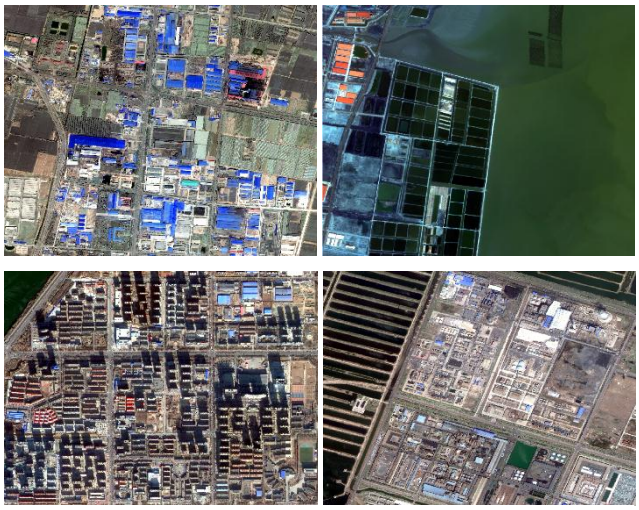


Figure 2. Sample image.

Rectangular templates are set to a different color to distinguish the marked artificial buildings, because most artificial buildings are annotation with a rectangle, the default shape when editing the drawing is set to a rectangle. After the vector layer property setting is completed, the screenshot is as shown in Figure 3.

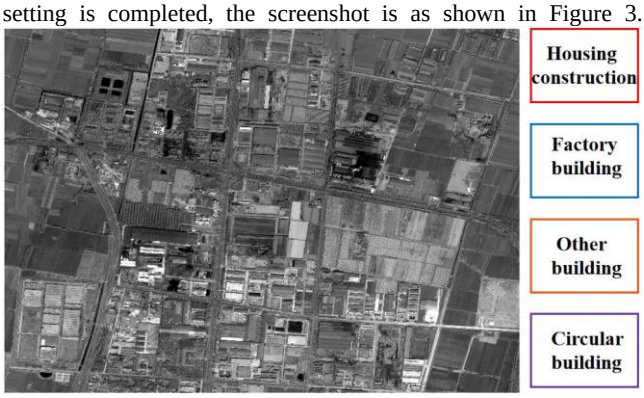


Figure 3. Property settings complete.

3.2.3 *Determine the category of annotation objects*

The principle of choosing annotation objects is to select artificial buildings with clear edges, clear categories and no occlusion, which are beneficial to the subsequent target detection, classification detection and other experiments.

Because the color of the panchromatic image is single, the building category can be decided by means of multispectral images or the surrounding environment when judging the artificial building category. Objects that are undetectable by the naked eye or are too small were not to annotation to ensure the accuracy of the labeled information.

Housing constructions are generally of high floor height, with long shadows, similar patterns and dense arrangement rules in the same area, and irregular roof shapes and even distribution rules. Factory buildings are generally low-rise buildings with large areas and rectangular roofs. Other buildings are buildings such as schools, stations and greenhouses, cannot be classified into specific categories. Circular buildings are better judged as regular domes.

Different colors of rectangles are used to mark different categories for easy distinction when labeling. In this paper, housing constructions are labeled in red, factory buildings are labeled in blue, circular buildings are labeled in purple, and other buildings are labeled in orange. The annotation frame must fit the target as much as possible when labeling target buildings, because it will correspond to multispectral images and fused images later, and rough labeling will have adverse effects on target detection, classification detection [10] and other experiments.

3.2.4 *Check the annotation and save*

After all the objects on the image have been labeled, whether the annotation information is in the attribute table must be checked, then delete the small template in the initial blank space and save the document in a relative path. Annotation information can be mapped to the multispectral and fused images. After labeling, a part of the property sheet is shown in table 1. In figure 4, we show the annotation results on multispectral images, panchromatic images and fused images.

Table 1. A part of the property sheet.

FID	Shape	Id	type
0	Polygon	0	Housing construction
1	Polygon	0	Factory building
2	Polygon	0	Circular building
3	Polygon	0	Other building
4	Polygon	0	Factory building
5	Polygon	0	Housing construction

3.2.5 *Summary*

Different types of images have their own advantages. For example, the panchromatic image has the highest resolution and the image is the clearest. The pseudo-color image is more conducive to observing the target environment. When people conduct scientific research, they may need according to different experimental requirements choose labeled data for multiple images. The efficiency of the annotation method proposed in this paper is not only reflected in the rigorous and meticulous setting of various attributes, but also reflected in the multiple use of one label. The annotation information is saved in the shp file. The geographic coordinates are the same so when panchromatic images, multispectral images or fused images need to be used, we only need to load the shp file and the image at the same time, which can save time and effort.

4. EXPERIMENT

Target detection experiments on the labeled samples using the Mask-RCNN and Faster-RCNN deep neural network model have been performed to prove and test that our labeled method is available and effective, which including 2692 training set images and 461 test set images with the size of 512*512 pixels.

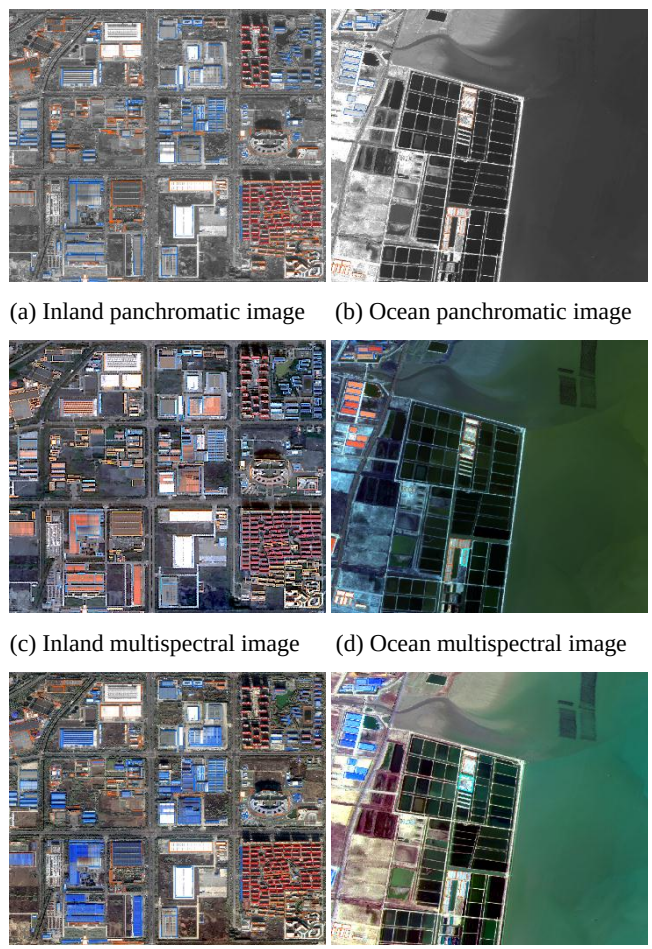
Loss function of the Mask-RCNN is as follows.

$$L = L_{cls} + L_{box} + L_{mask} \tag{1}$$

In this formula, L_{cls} represents classification error, L_{box}represents detection error, and L_{mask} represents segmentation error.

Some basic parameters of this target detection experiment using Mask-RCNN are set as conv_body=R-50-FPN, anchor_sizes=(8, 16, 32, 64, 128), score_thresh=0.5, base_lr=0.005, weight_decay=0.0001, min_size_train=(512, 576, 640, 704, 800), min_size_test=800.

Experimental results of target detection using Mask-RCNN are shown in table 2.



(a) Inland panchromatic image (b) Ocean panchromatic image

(c) Inland multispectral image (d) Ocean multispectral image

(e) Inland fused image (f) Ocean fused image

Figure 4. Labeled completion figure.

Loss function of Faster-RCNN is as follows.

$$L(\{P_i\},\{t_i\}) = \frac{1}{N_{cls}}\sum_i L_{cls}(p_i, p_i^*) + \lambda\frac{1}{N_{reg}}\sum_i p_i^* L_{reg}(t_i, t_i^*) \quad (2)$$

P_i is the probability that the anchor prediction is the target. p_i^* is GT label. t_i is 4 parameterized coordinates of the predicted bounding box, t_i^* is the coordinate vector of the ground truth bounding box corresponding to a positive anchor.

Some basic parameters of this target detection experiment using Faster-RCNN are set as conv_body=R-50-FPN, anchor_sizes=(8, 16, 32, 64, 128)，score_thresh=0.5, base_lr=0.1，weight_decay=0.0001，min_size_train=(512, 576, 640, 704, 800)，min_size_test=800。

Experimental results of target detection using Faster-RCNN are shown in table 3.

The usability of this annotation method can be proved.

Table 2. Mask-RCNN experimental results.

IoU	area	maxDets	Average Precision (AP)	Average Recall (AR)
0.50:0.95	all	100	0.232	0.368
0.50	all	100	0.473	--
0.75	all	100	0.208	--
0.50:0.95	small	100	0.142	0.286
0.50:0.95	medium	100	0.390	0.526
0.50:0.95	large	100	0.476	0.554
0.50:0.95	all	1	--	0.029
0.50:0.95	all	10	--	0.190

Table 3. Faster-RCNN experimental results.

IoU	area	maxDets	Average Precision (AP)	Average Recall (AR)
0.50:0.95	all	100	0.260	0.388
0.50	all	100	0.508	--
0.75	all	100	0.238	--
0.50:0.95	small	100	0.183	0.300
0.50:0.95	medium	100	0.403	0.556
0.50:0.95	large	100	0.471	0.632
0.50:0.95	all	1	--	0.030
0.50:0.95	all	10	--	0.206

5. CONCLUSION

This paper proposed an efficient annotation method for large scale remote sensing dataset. The traditional annotation flowchart was improved via parallel labeling method. The proposed method plays a very important role in the labeling of GF-Buildings dataset, which shortens the labeling time and improves the accuracy of labeling. In the experiment of remote sensing target detection, the effectiveness of the proposed method is verified by using Mask-RCNN and Faster-RCNN, and achieved satisfactory experimental results.

6. ACKNOWLEDGMENTS

This work is supported by Science and Technology Development Program of Jinan, Natural Science Foundation of University of Jinan (No. XKY1804, XKY1803, XKY1928) and A Teaching Cases Project of Postgraduate Education in Shandong Province (Grant NO. SDYAL18028).

7. REFERENCES

[1] Mishra, V. N., Prasad, R., Rai, P. K., Vishwakarma, A. K., and Arora, A. 2019. Performance evaluation of textural features in improving land use/land cover classification accuracy of heterogeneous landscape using multi-sensor remote sensing data. Earth Science Informatics, 12,1, 71-86.

[2] Chen, J., Liu, H., Hou, J., Yang, M., and Deng, M. 2018. Improving Building Change Detection in VHR Remote Sensing Imagery by Combining Coarse Location and Co-Segmentation. ISPRS International Journal of Geo-Information, 7,6, 213.

[3] Zhang, Q., Wang, Y., Liu, Q., Liu, X., and Wang, W. 2016. CNN based suburban building detection using monocular high resolution Google Earth images. In 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS). (July. 2016), pp. 661-664. IEEE.

[4] Vuola, A. O., Akram, S. U., and Kannala, J. 2019. Mask-RCNN and U-net Ensembled for Nuclei Segmentation. arXiv preprint arXiv:1901.10170.

[5] Zhao, B., Zhong, Y., Xia, G. S., and Zhang, L. 2015. Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery. IEEE Transactions on Geoscience and Remote Sensing, 54,4, 2108-2123.

[6] Luus, F. P., Salmon, B. P., Van den Bergh, F., and Maharaj, B. T. J. 2015. Multiview deep learning for land-use classification. IEEE Geoscience and Remote Sensing Letters, 12,12, 2448-2452.

[7] Shen, Y., Zhou, X., Liu, J., and Chen, J. 2018. CSRS-SIAT: A benchmark remote sensing dataset to semantic-enabled and cross-scales scene recognition. In IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium. (July. 2018), pp. 1780-1783. IEEE.

[8] Zhu, Q., Zhong, Y., Zhao, B., Xia, G. S., and Zhang, L. 2016. Bag-of-visual-words scene classifier with local and global features for high spatial resolution remote sensing imagery. IEEE Geoscience and Remote Sensing Letters, 13,6, 747-751.

[9] Lin, T. Y., Dollár, P., Girshick, R., He, K., Hariharan, B., and Belongie, S. 2017. Feature pyramid networks for object detection. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2117-2125.

[10] Ma, M., Chen, J., Liu, W., & Yang, W. 2018. Ship Classification and Detection Based on CNN Using GF-3 SAR Images. Remote Sensing, 10,12, 2043